WORKED SOLUTION

# STEP II 2000, Question 3

Pure • Error Propagation

**STEP II 2000, Question 3**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q3

## The key idea

The cosine rule gives $a^2$ exactly in terms of $b$, $c$ and $A$, so the whole question is about how $a$ responds when each of these is nudged slightly. Expand every ingredient to first order in the small quantities, throw away products of small quantities, and finally take a square root using $\sqrt{1+x} \approx 1 + \frac{1}{2}x$. For the range, notice that $\eta$ is a linear combination of independent errors, so it is largest when each term is pushed to its extreme with the helpful sign.

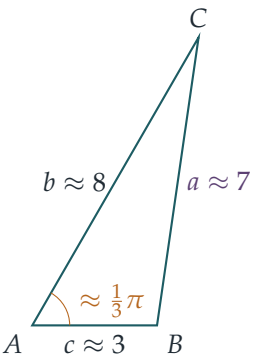
## Solution

### The approximation for $a$

With the unperturbed values $b = 8$, $c = 3$, $A = \frac{1}{3}\pi$, the cosine rule gives

$$a^2 = 64 + 9 - 2 \cdot 8 \cdot 3 \cdot \tfrac{1}{2} = 49,$$

so $a = 7$. This explains the leading term.




Now perturb. Throughout, "≈" means we keep terms of first order in $\epsilon_1, \epsilon_2, \epsilon_3$ and

discard anything quadratic or higher, such as $\epsilon_1^2$ or $\epsilon_2\epsilon_3$. Then

$$\begin{aligned} b^2 &= 64 + 16\epsilon_1 + \epsilon_1^2 \approx 64 + 16\epsilon_1, \\ c^2 &= 9 + 6\epsilon_2 + \epsilon_2^2 \approx 9 + 6\epsilon_2, \\ bc &= 24 + 3\epsilon_1 + 8\epsilon_2 + \epsilon_1\epsilon_2 \approx 24 + 3\epsilon_1 + 8\epsilon_2. \end{aligned}$$

For the angle, the compound-angle formula and the small-angle approximations $\cos\epsilon_3 \approx 1$, $\sin\epsilon_3 \approx \epsilon_3$ give

$$\cos A = \cos\tfrac{1}{3}\pi\cos\epsilon_3 - \sin\tfrac{1}{3}\pi\sin\epsilon_3 \approx \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\epsilon_3.$$

(The error in $\cos\epsilon_3 \approx 1$ is of order $\epsilon_3^2$, so it really is negligible at this order.) Multiplying out and again dropping products of small quantities,

$$2bc\cos A \approx 2(24 + 3\epsilon_1 + 8\epsilon_2)\left(\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\epsilon_3\right) \approx 24 + 3\epsilon_1 + 8\epsilon_2 - 24\sqrt{3}\,\epsilon_3.$$

Putting the pieces into the cosine rule,

$$\begin{aligned} a^2 &\approx (64 + 16\epsilon_1) + (9 + 6\epsilon_2) - (24 + 3\epsilon_1 + 8\epsilon_2 - 24\sqrt{3}\,\epsilon_3) \\ &= 49 + 13\epsilon_1 - 2\epsilon_2 + 24\sqrt{3}\,\epsilon_3. \end{aligned}$$

Write $\delta = 13\epsilon_1 - 2\epsilon_2 + 24\sqrt{3}\,\epsilon_3$, which is small. Since $a > 0$,

$$a = \sqrt{49 + \delta} = 7\left(1 + \frac{\delta}{49}\right)^{1/2} \approx 7\left(1 + \frac{\delta}{98}\right) = 7 + \frac{\delta}{14},$$

using the binomial expansion $(1 + x)^{1/2} \approx 1 + \frac{1}{2}x$ for small $x$. Hence $a \approx 7 + \eta$ with

$$\eta = \frac{13\epsilon_1 - 2\epsilon_2 + 24\sqrt{3}\,\epsilon_3}{14},$$

which is the required result.

**Ask yourself** Could we have avoided expanding products by hand?

Yes: differentiate $a^2 = b^2 + c^2 - 2bc\cos A$ to get $2a\,\delta a = (2b - 2c\cos A)\,\delta b + (2c - 2b\cos A)\,\delta c + 2bc\sin A\,\delta A$. At $a = 7$, $b = 8$, $c = 3$, $A = \frac{1}{3}\pi$ this reads $14\,\delta a = 13\,\delta b - 2\,\delta c + 24\sqrt{3}\,\delta A$, the same answer in one line. Both methods are just first-order Taylor expansion.

### The range of $\eta$

The bound on $\epsilon_2$ is printed in the paper as $4\cdot 9 \times 10^{-2}$, with the old-style raised decimal point, so it means $|\epsilon_2| \le 4.9 \times 10^{-2}$.

The three errors vary independently, so by the triangle inequality

$$|\eta| \le \frac{13|\epsilon_1| + 2|\epsilon_2| + 24\sqrt{3}\,|\epsilon_3|}{14} \le \frac{13 \times 2 \times 10^{-3} + 2 \times 4.9 \times 10^{-2} + 24\sqrt{3} \times \sqrt{3} \times 10^{-3}}{14}.$$

The numerator is $0.026 + 0.098 + 0.072 = 0.196$, so

$$|\eta| \leq \frac{0.196}{14} = 0.014.$$

The bound is attained: taking $\epsilon_1 = 2 \times 10^{-3}$, $\epsilon_2 = -4.9 \times 10^{-2}$, $\epsilon_3 = \sqrt{3} \times 10^{-3}$ makes every term of $\eta$ positive and as large as possible, giving $\eta = 0.014$; reversing all three signs gives $\eta = -0.014$. Since $\eta$ is a continuous (indeed linear) function of the errors, and the set of allowed errors is a box, every value in between is also taken. So the range is

$$-0.014 \leq \eta \leq 0.014,$$

that is, $a$ lies in roughly $7 \pm 0.014$.

## Checks

- With all errors zero the triangle is 8, 3, 7 with a $60^\circ$ angle, and indeed $64 + 9 - 24 = 49$.
- Taking the extreme errors above, the exact cosine rule gives $a \approx 7.0141$ and $a \approx 6.9861$, very close to $7 \pm 0.014$; the small discrepancy is the discarded second-order terms.
- Numerically, the partial derivative of $a$ with respect to $A$ at the base values is $2.9692\ldots = 24\sqrt{3}/14$, as predicted.

## Common slips

- Forgetting the square root at the end: $a^2 \approx 49 + \delta$ gives $a \approx 7 + \delta/14$, not $7 + \delta$.
- Taking all three errors positive for the maximum: $\epsilon_2$ enters with a minus sign, so it must be negative to push $\eta$ up.
- Reading $4 \cdot 9 \times 10^{-2}$ as $36 \times 10^{-2}$; that is not a small error at all for a side of length 3.

## Beyond STEP

This is error propagation through the total differential: for a function $f$ of several variables, $\delta f \approx \sum (\partial f / \partial x_i)\, \delta x_i$, and the worst case is $\sum |\partial f / \partial x_i|\, |\delta x_i|$. In a first-year course this becomes the multivariable Taylor theorem, with an explicit bound on the second-order remainder that tells you exactly how good "$\approx$" is. Experimental scientists often use instead the root-sum-of-squares rule for independent random errors. Bartle and Sherbert, *Introduction to Real Analysis*, covers Taylor's theorem with remainder carefully.

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